Data-Backed Journal-Style Replication of *Knowledge Networks as Channels and Conduits: The Effects of Spillovers in the Boston Biotechnology Community*

Replication Paper Project

2026-04-07

Abstract

This paper presents a methodology-specific local analogue of Knowledge Networks as Channels and Conduits: The Effects of Spillovers in the Boston Biotechnology Community using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 2,378 observations across 177 entities, and the batch headline estimate is 0.00085 for Component Size x Open Science with p = 0.18. The local question is whether open-science orientation shifts the innovation benefits of network membership away from narrow centrality and toward broader component exposure. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: knowledge networks; spillovers; openness; component membership; centrality; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of Knowledge Networks as Channels and Conduits: The Effects of Spillovers in the Boston Biotechnology Community and treats the source article as

an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks whether network benefits accrue through centrality alone or also through broader component membership when ties are embedded in more open spillover regimes (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication matters because network papers often treat all ties as equivalent even though openness can change what a tie actually delivers. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 2,378 observations across 177 entities and spans 1991 to 2014 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Owen-Smith and Powell, 2004; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent openness-by-network-position analogue that can be estimated from the current citation-network and science panels. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Owen-Smith and Powell, 2004; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper sits in a literature on knowledge networks, spillovers, openness, and the conditions under which network position creates broader or narrower innovation benefits. (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the local paper is not only about whether ties matter, but about what kind of network access matters under more open knowledge regimes. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja

and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). The review therefore adds network and spillover work that sharpens the difference between centrality and broader component exposure. (Rosenkopf and Almeida, 2003; Ahuja, 2000; Stuart and Podolny, 1996). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Schilling and Phelps, 2007; Fleming et al., 2007). A final topic-specific literature block keeps the bibliography tied to the paper's actual mechanism rather than to a generic replication template. (Ahuja and Katila, 2001; Lavie et al., 2011; Griliches, 1990).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Owen-Smith and Powell, 2004). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper contrasts channels and conduits as different ways network spillovers may travel. Geography and organizational form alter how accessible network spillovers are in the source argument. The empirical design distinguishes the value of centrality from the value of broader network membership. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the

local executable analogue (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Owen-Smith and Powell (2004) starts with the published empirical design rather than with the local rerun. The source article studies how geographic propinquity and organizational form alter the accessibility of spillovers by building yearly Boston biotechnology networks from formal interorganizational ties (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published sample is described as follows: The published sample is a set of human therapeutic biotechnology firms located in the Boston metropolitan area and observed through yearly network snapshots. The published measurement architecture is described as follows: The published design constructs Boston and Boston-plus biotechnology networks, distinguishes main-component membership from betweenness centrality, and links those positions to patenting. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published estimator is described as follows: conditional fixed-effects negative binomial regressions of successful patent applications for human therapeutic biotechnology firms in the Boston metropolitan area. The central identifying comparison is whether innovation benefits accrue to betweenness centrality or to broader network-component membership under different openness conditions, net of firm and year effects. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). This paper is a methodology-specific local analogue because the original article is built on Boston-specific biotechnology networks and conditional fixed-effects negative binomial models, whereas the local rerun uses a broad public-firm citation-link network with open-science proxies. The present manuscript can test the channels-versus-conduits logic, but not recreate the original regional network archive (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper distinguishes centrality from broader component membership. The local reconstruction therefore adopts the following bounded decision: Direct ties and component size carry the centrality-versus-membership distinction in the local network. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Openness changes spillover accessibility in the source argument. The local reconstruction therefore adopts the following bounded decision: Open science is proxied with publication and public-research exposure in the current data stack. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw

archival pipeline (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Network spillovers are the mechanism of interest. The local reconstruction therefore adopts the following bounded decision: Next-year patenting is used as the executable performance outcome. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Rolling firm-year interfirm citation-link network derived from processed backward patent ties. Firm-year open-science indicator built from publications and public-research citation counts already present in the science panel. Firm-year patent stock, scale, and R&D controls from the merged public-firm panel (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 2,378 observations across 177 entities and covers 1991 to 2014 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending

that the original archive has been fully recreated (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue uses a rolling interfirm citation-link network, a component-size measure, and an open-science indicator built from publications and public-research citations to distinguish centrality from broader network access. The local sample frame is Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes The observed time window in the executable panel is 1991 to 2014 in the local executable panel The outcome is Next-year patent count at the firm-year level (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Owen-Smith and Powell, 2004; Hall et al., 2005).

3.1 Local Data Sources Used in This Replication

The current paper is estimated from a defined local data stack rather than from an unnamed replication pool. The executable evidence chain uses the following data objects: analysis panel, coefficient export, sample summary, fit summary. The body of the paper refers to these objects by analytic role rather than by raw filename so that the data section reads like a paper instead of a file dump (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). Those objects are not interchangeable: analysis panel supports estimate the local model and regenerate descriptive statistics; coefficient export supports recover the reported local coefficients and p-values; sample summary supports audit sample construction and raw grouped moments; fit summary supports check model diagnostics when a fit file is available. Together, they support 2,378 observations across 177 entities over 1991 to 2014 in the local executable panel (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). This local data stack is narrower than the original article's data architecture. Rolling firm-year interfirm citation-link network derived from processed backward patent ties. Firm-year open-science indicator built from publications and public-research citation counts already present in the science panel. Firm-year patent stock, scale, and R&D controls from the merged public-firm panel. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the local processed data objects supply the estimable analogue. Exact filenames are retained in the replication notes rather than printed

through the body text (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005).

DataObject	RoleInPaper
Analysis panel	Estimate the local model and regenerate descriptive statistics
Coefficient export	Recover the reported local coefficients and p-values
Sample summary	Audit sample construction and raw grouped moments
Fit summary	Check model diagnostics when a fit file is available

Local data objects used in the present paper.

3.2 Sample Reconstruction

The local analysis file contains 2,378 observations drawn from 177 entities. The observed time span runs from 1991 to 2014. Selected sample moments are Next Patents = 228.2, Direct Ties = 37.08, Component Size = 98.04, Open Science = 3.440 (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	2,378
Unique Firms	177
Start Year	1991
End Year	2014
Mean Next Patents	228
Mean Direct Ties	37.08

Statistic	Value
Mean Component Size	98.04
Mean Open Science	3.440

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Centrality is operationalized locally as Direct ties and appears in the executable dataset as direct_ties. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use degree in the five-year rolling citation-link network.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Broader component membership is operationalized locally as Connected-component size and appears in the executable dataset as component_size. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Compute the size of the focal firm's connected component in that network.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart

and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Openness is operationalized locally as Open-science indicator and appears in the executable dataset as open_science. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Take log(1 + publications + external public-research citations + internal science citations).. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Innovation outcome is operationalized locally as Next-year patent count and appears in the executable dataset as next_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Lead annual patent counts by one year within firm.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Firm-year controls is operationalized locally as Patent stock, assets, R&D intensity and appears in the executable dataset as log_pat_stock; log_assets; rd_intensity. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the already merged firm-year controls.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a

black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Centrality	Direct ties	Use degree in the five-year rolling citation-link network.
Broader component membership	Connected-component size	Compute the size of the focal firm's connected component in that network.
Openness	Open-science indicator	Take log(1 + publications + external public-research citations + internal science citations).
Innovation outcome	Next-year patent count	Lead annual patent counts by one year within firm.
Firm-year controls	Patent stock, assets, R&D intensity	Reuse the already merged firm-year controls.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Owen-Smith and Powell (2004). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Centrality, the manuscript uses Direct ties under the exact local implementation direct_ties. The derivation step is: Use degree in the five-year rolling citation-link network.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Broader component membership, the manuscript uses Connected-component size under the exact local implementation component_size. The derivation step is: Compute the size of the focal firm's connected component in that network.. This construct is retained because it occupies the same analytic position in the local design that the published paper

assigns to the original concept. For Openness, the manuscript uses Open-science indicator under the exact local implementation open_science. The derivation step is: Take log(1 + publications + external public-research citations + internal science citations).. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Innovation outcome, the manuscript uses Next-year patent count under the exact local implementation next_patents. The derivation step is: Lead annual patent counts by one year within firm.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm-year controls, the manuscript uses Patent stock, assets, R&D intensity under the exact local implementation log_pat_stock; log_assets; rd_intensity. The derivation step is: Reuse the already merged firm-year controls.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

For a channels-versus-conduits paper, the descriptive section needs to map the architecture of the network before it talks about patenting outcomes. For Next-year patents, the mean is 228.2, the standard deviation is 519.3, and the interquartile range runs from 20.00 to 226.4; For Direct ties, the mean is 37.08, the standard deviation is 22.75, and the interquartile range runs from 18.00 to 53.00; For Component size, the mean is 98.04, the standard deviation is 27.99, and the interquartile range runs from 81.00 to 122.0; For Open science, the mean is 3.440, the standard deviation is 2.215, and the interquartile range runs from 1.609 to 5.329 . The first descriptive task is to show whether direct ties, component size, and open science occupy different empirical roles in the local network rather than acting as different names for the same position variable (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time profile is read as a history of network thickening and openness. That matters because the source theory is about when spillovers stay narrow and when they diffuse through broader components. the value of firm-year observations is 132 in 1991 and 75 in 2014; the value of mean next-year patents is 118.0 in 1991 and 462.9 in 2014; the value of mean direct ties is 23.27 in 1991 and 36.11

in 2014 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The nonparametric support check is interpreted as an openness gradient, asking whether innovation outcomes shift as firms move across the support of open-science exposure in the networked panel. Mean outcomes range from 19.78 to 660.1 across the support of the focal regressor, while the mean regressor itself ranges from 0.384 to 6.519. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation matrix is used to distinguish narrow centrality from broader connectedness. If direct ties and component size were almost perfectly aligned, the conduit-versus-channel contrast would disappear in the local data. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Setting-Specific Figure Evidence

The figure is meant to behave like a network-structure figure rather than like a generic panel plot (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The left panel keeps direct ties, component size, and open-science exposure in view simultaneously, while the right panel shows how next-year patenting changes across the openness support that matters for the conduit-versus-channel comparison. mean direct ties moves from 23.27 in 1991 to 36.11 in 2014; mean component size moves from 111.9 in 1991 to 66.33 in 2014; mean open-science exposure moves from 2.817 in 1991 to 4.155 in 2014 Across the support panel, mean outcome moves from 19.78 to 660.1 as the mean regressor shifts from 0.384 to

6.519 (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

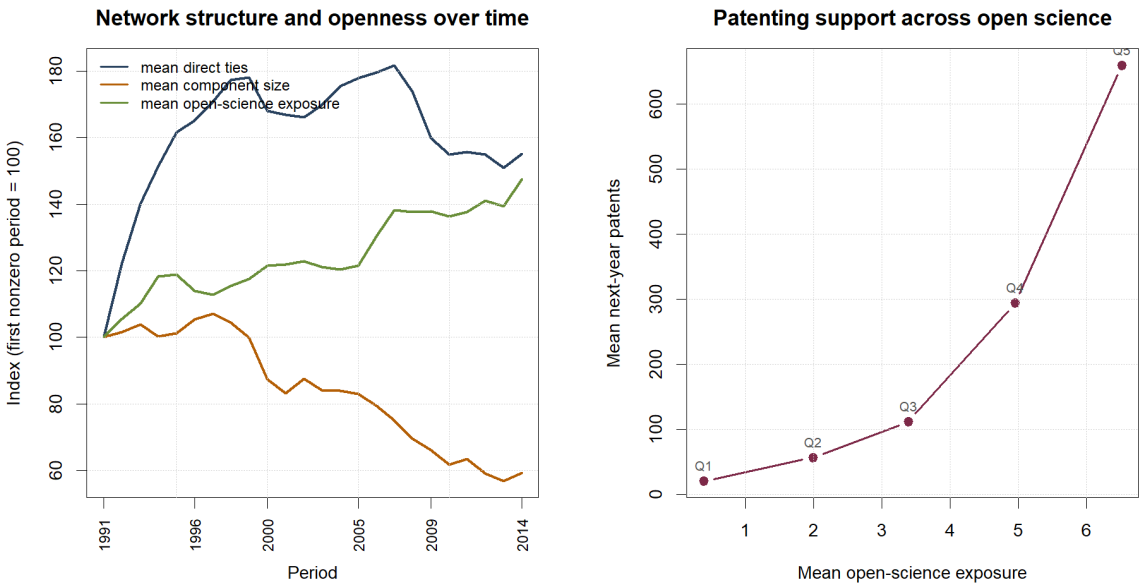

Figure 1: Figure 1. Network architecture, open-science exposure, and the observed patenting support across the openness gradient.

4.2 Network Coverage and Structural Support

For this paper, data quality is about network structure as much as it is about missingness. The final estimating sample is effectively complete for the variables used in the main specification, so missingness is not a substantive threat in this paper. For that reason, the manuscript does not spend body-table space on an all-zero missingness display (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). Coverage is reported to show whether the network result reflects broad component membership or a few firms that dominate the citation-link graph. The most heavily represented entities contribute between 24 and 24 observations each. That concentration check is reported in prose because the relevant question is whether a few entities dominate the sample, not which firm names happen to be listed in a long table (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). Support is emphasized because the conduit-versus-channel argument depends on firms occupying meaningfully different positions in the openness gradient. In the lowest observed quantile, the mean regressor is 0.384 and the mean outcome is 19.78 across 484 observations. In the highest observed quantile, the corresponding values are 6.519, 660.1, and 476 observations. This

shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). These diagnostics matter because a network paper can become trivial if centrality, connectedness, and openness collapse into the same empirical signal. The section is therefore designed to keep those roles distinct before the model is interpreted (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005).

Period	FirmYears	MeanNextPatents	MeanDirectTies	MeanComponentSize	MeanOpenScience
1991	132	118	23.27	111.9	2.817
1992	130	122	28.52	113.6	2.972
1993	126	130	32.61	116.2	3.103
1994	123	129	35.21	112.3	3.332
1995	122	140	37.60	113.2	3.351
1996	123	130	38.45	118.1	3.209
1997	120	170	39.77	120.0	3.179
1998	113	183	41.24	116.9	3.251
1999	106	195	41.42	111.9	3.310
2000	102	209	39.10	97.76	3.422
2001	100	219	38.85	93.15	3.435
2002	98	226	38.65	98.00	3.464
2003	100	216	39.50	94.12	3.412
2004	98	195	40.84	94.04	3.392
2005	93	250	41.40	92.98	3.425
2006	89	232	41.79	88.93	3.674
2007	83	258	42.29	83.89	3.892
2008	78	292	40.42	77.92	3.878
2009	75	387	37.21	73.96	3.884
2010	75	374	36.07	69.16	3.840
2011	72	425	36.21	71.00	3.876
2012	70	463	36.04	66.06	3.974
2013	75	468	35.12	63.56	3.927
2014	75	463	36.11	66.33	4.155

Observed time patterns relevant to the focal empirical setting.

Variable	Next-year patents	Direct ties	Component size	Open science	Log patent stock	Log assets	R&D intensity	External public-research citations
Next-year patents	1	0.419	-0.064	0.480	0.476	0.436	0.113	0.814
Direct ties	0	1.000	0.227	0.653	0.774	0.567	0.054	0.242
Component size	0	0.227	1.000	0.089	0.263	-0.148	0.037	-0.100
Open science	0	0.653	0.089	1.000	0.757	0.586	0.381	0.434
Log patent stock	0	0.774	0.263	0.757	1.000	0.536	0.158	0.323
Log assets	0	0.567	-0.148	0.586	0.536	1.000	-0.138	0.307

Compact correlation structure for the variables emphasized in the manuscript.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A firm and year fixed-effects count model of next-year patenting on direct ties, component size, open science, and the interactions of openness with centrality and component membership. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Owen-Smith and Powell, 2004; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes, and the empirical window is 1991 to 2014 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal

and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Next-year patent count at the firm-year level, and the main explanatory design is Firm and year fixed-effects count model with direct ties, component size, open science, and openness interactions. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Patent stock, assets, and R&D intensity plus firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps only firm-years supported jointly by the rolling citation-link network and the merged public-firm science panel.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes over 1991 to 2014 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say

about the mechanism under study. Measurement decisions: the outcome is Next-year patent count at the firm-year level and the main explanatory design is Firm and year fixed-effects count model with direct ties, component size, open science, and openness interactions. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Patent stock, assets, and R&D intensity plus firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps only firm-years supported jointly by the rolling citation-link network and the merged public-firm science panel. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$E[\text{NextPatents}_{it+1} \mid X_{it}] = \exp\left(\beta_1 \text{DirectTies}_{it} + \beta_2 \text{ComponentSize}_{it} + \beta_3 \text{OpenScience}_{it} + \beta_4 \text{DirectTies}_{it} \times \text{OpenScience}_{it} + \beta_5 \text{ComponentSize}_{it} \times \text{OpenScience}_{it} + \beta_6 X_{it} + \alpha_i + \gamma_t\right). \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is 0.00085 on Component Size x Open Science (p = 0.18). The local question is whether open-science orientation shifts the innovation benefits of network membership away from narrow centrality and toward broader component exposure. Log patent stock is estimated at 0.764 with p = ¡ .001. Log assets is estimated at 0.244 with p = ¡ .001. R&D intensity is estimated at 4.097 with p = ¡ .001 (Owen-Smith and Powell, 2004; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
Direct ties	-0.0062	0.0054	0.246
Component size	-0.0040	0.0021	0.0604
Open science	-0.0532	0.0826	0.519
Log patent stock	0.7644	0.0758	6.11e-24
Log assets	0.2436	0.0714	0.0006
R&D intensity	4.097	1.240	0.0010

Term	Estimate	Std. error	P-value
Direct ties x open science	-0.0007	0.0007	0.375
Component size x open science	0.0008	0.0006	0.185

Key coefficient table used in the main text.

6.2 Visual Summary of the Main Estimates

The coefficient figure makes it easier to compare direct ties, component size, open science, and their interactions without asking the reader to decode the network argument from rows of numbers alone (Owen-Smith and Powell, 2004; Hall et al., 2005).

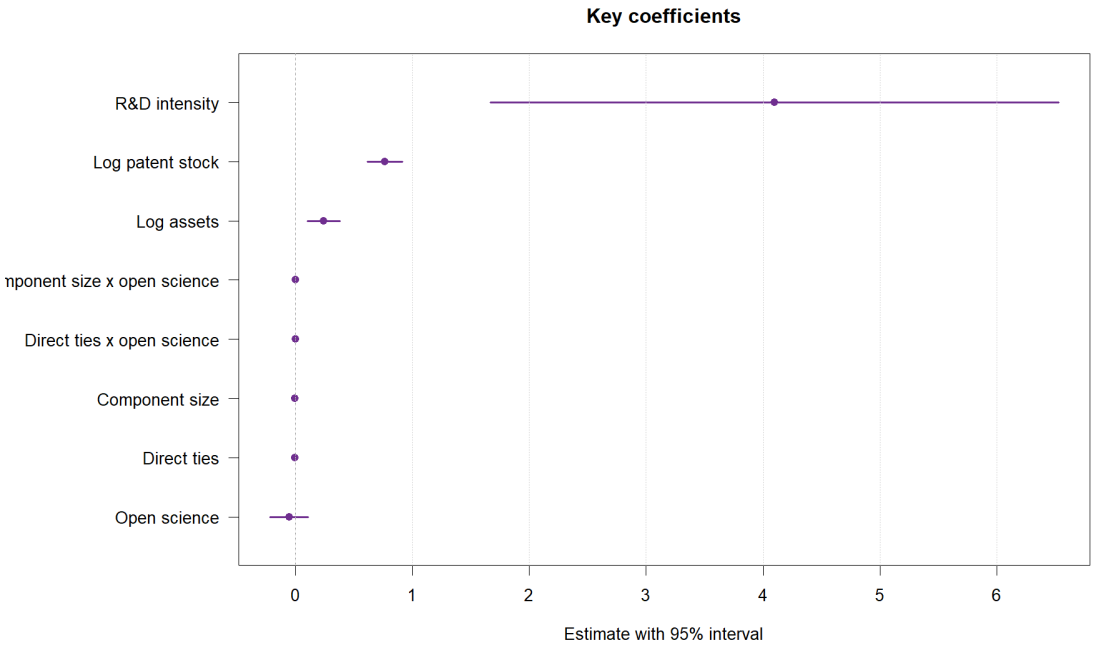

Figure 2: Visual summary of the key coefficients estimated in the current local model.

6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log patent stock is estimated at 0.764 with a p-value of ¡ .001; Log assets is estimated at 0.244 with a p-value of ¡ .001; R&D intensity is estimated at 4.097 with a p-value of ¡ .001; Component size is estimated at -0.0040 with a p-value of 0.060 (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Log patent stock, which is estimated at 0.764 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Next-year patents has mean 228.2 and standard deviation 519.3 in the local panel, while the main regressor Open science has mean 3.440 and standard deviation 2.215 (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 19.78 to 660.1 while the mean regressor ranges from 0.384 to 6.519. This does not substitute for the model, but it shows how much raw variation the specification must explain (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local question is whether open-science orientation shifts the innovation benefits of network membership away from narrow centrality and toward broader component exposure. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within baseline, the leading coefficient family reads as follows: Log patent stock = 0.764 (p = ¡ .001); Log assets = 0.244 (p = ¡ .001); R&D intensity = 4.097 (p = ¡ .001); Component size = -0.0040 (p = 0.060); Component size x open science = 0.00085 (p = 0.18). The largest positive estimate in this specification is R&D intensity at 4.097, while the largest negative estimate is Open science at -0.0532. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Owen-Smith and Powell (2004) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Owen-Smith and Powell, 2004; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Log patent stock with estimate 0.764 and p = ¡ .001 (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Baseline specification contributes its own reading to the replication. Log patent stock is estimated at 0.764 with p = ¡ .001 and is classified as a strong signal; Log assets is

estimated at 0.244 with p = ¡ .001 and is classified as a strong signal; R&D intensity is estimated at 4.097 with p = ¡ .001 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Owen-Smith and Powell (2004) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

6.6 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). For knowledge_networks, the current fit export reports N = 2,276, pseudo R-squared = 0.953, R-squared = NA (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Owen-Smith and Powell (2004) is being carried out on a clearly described estimating sample with observable fit statistics (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005).

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Owen-Smith and Powell, 2004). For Owen-Smith and Powell (2004), the local headline result is Component Size x Open Science = 0.00085 with p = 0.18. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper finds that the accessibility of spillovers depends on openness and that benefits can accrue to broad network membership rather than only to centrality.	Component Size x Open Science = 0.00085 with p = 0.18.	The published paper argues that openness changes whether innovation benefits accrue narrowly to central firms or more broadly to firms embedded in a coherent network component. The local analogue translates that question into a citation-link network with a component-size measure and an open-science proxy.

Aspect	OriginalPaper	LocalDataset	Reading
Measurement and data construction	The published design constructs Boston and Boston-plus biotechnology networks, distinguishes main-component membership from betweenness centrality, and links those positions to patenting.	The local analogue uses a rolling interfirm citation-link network, a component-size measure, and an open-science indicator built from publications and public-research citations to distinguish centrality from broader network access.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published sample is a set of human therapeutic biotechnology firms located in the Boston metropolitan area and observed through yearly network snapshots.	Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes. Local sample = 2,378 observations across 177 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design constructs Boston and Boston-plus biotechnology networks, distinguishes main-component membership from betweenness centrality, and links those positions to patenting By contrast, the local paper relies on the following executable design: The local analogue uses a rolling interfirm citation-link network, a component-size measure, and an open-science indicator built from

publications and public-research citations to distinguish centrality from broader network access These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published sample is a set of human therapeutic biotechnology firms located in the Boston metropolitan area and observed through yearly network snapshots The local sample instead works from Firm-year panel with rolling citation-network measures, open-science indicators, and next-year patenting outcomes and yields 2,378 observations across 177 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The published paper argues that openness changes whether innovation benefits accrue narrowly to central firms or more broadly to firms embedded in a coherent network component. The local analogue translates that question into a citation-link network with a component-size measure and an open-science proxy. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The published paper argues that openness changes whether innovation benefits accrue narrowly to central firms or more broadly to firms embedded in a coherent network component. The local analogue translates that question into a citation-link network with a component-size measure and an open-science

proxy. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

The published paper argues that openness changes whether innovation benefits accrue narrowly to central firms or more broadly to firms embedded in a coherent network component. The local analogue translates that question into a citation-link network with a component-size measure and an open-science proxy (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Owen-Smith and Powell, 2004; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The published paper studies the Boston biotechnology community and includes richer geography and organizational-form structure (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. The local rerun uses a broad public-firm citation-link network and an open-science proxy in place of the original regional network design (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The paper is therefore strongest as a transportability test of

the channels-versus-conduits logic (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent openness-by-network-position analogue that can be estimated from the current citation-network and science panels (Owen-Smith and Powell, 2004; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Owen-Smith and Powell, 2004; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	2,378	rows
Unique Firms	177	unique_firms
Start Year	1991	year_min
End Year	2014	year_max
Mean Next Patents	228	mean_next_patents
Mean Direct Ties	37.08	mean_direct_ties
Mean Component Size	98.04	mean_component_size
Mean Open Science	3.440	mean_open_science

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Next-year patents	2,378	228.2	519.3	20.00	78.50	226.4	0	7662
Direct ties	2,378	37.08	22.75	18.00	39.00	53.00	0	99
Component size	2,378	98.04	27.99	81.00	99.00	122.0	1	123
Open science	2,378	3.440	2.215	1.609	3.367	5.329	0	9
Log patent stock	2,378	5.803	2.097	4.833	6.254	7.236	0	10
Log assets	2,378	9.388	1.341	8.480	9.310	10.26	4	14
R&D intensity	2,378	0.0417	0.0403	0.0134	0.0281	0.0594	0	1
External public-research citations	2,378	47.66	177.7	0.0000	2.000	20.00	0	2721

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Next-year patents	Direct ties	Component size	Open science	Log patent stock	Log assets	R&D intensity	External public-research citations
Next-year patents	1	0.419	-0.064	0.480	0.476	0.436	0.113	0.814
Direct ties	0	1.000	0.227	0.653	0.774	0.567	0.054	0.242
Component size	0	0.227	1.000	0.089	0.263	-0.148	0.037	-0.100
Open science	0	0.653	0.089	1.000	0.757	0.586	0.381	0.434
Log patent stock	0	0.774	0.263	0.757	1.000	0.536	0.158	0.323
Log assets	0	0.567	-0.148	0.586	0.536	1.000	-0.138	0.307
R&D intensity	0	0.054	0.037	0.381	0.158	-0.138	1.000	0.139
External public-research citations	1	0.242	-0.100	0.434	0.323	0.307	0.139	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	FirmYears	MeanNextPatents	MeanDirectTies	MeanComponentSize	MeanOpenScience
1991	132	118	23.27	111.9	2.817
1992	130	122	28.52	113.6	2.972
1993	126	130	32.61	116.2	3.103
1994	123	129	35.21	112.3	3.332
1995	122	140	37.60	113.2	3.351
1996	123	130	38.45	118.1	3.209
1997	120	170	39.77	120.0	3.179
1998	113	183	41.24	116.9	3.251
1999	106	195	41.42	111.9	3.310
2000	102	209	39.10	97.76	3.422
2001	100	219	38.85	93.15	3.435
2002	98	226	38.65	98.00	3.464
2003	100	216	39.50	94.12	3.412
2004	98	195	40.84	94.04	3.392
2005	93	250	41.40	92.98	3.425
2006	89	232	41.79	88.93	3.674
2007	83	258	42.29	83.89	3.892
2008	78	292	40.42	77.92	3.878
2009	75	387	37.21	73.96	3.884
2010	75	374	36.07	69.16	3.840
2011	72	425	36.21	71.00	3.876
2012	70	463	36.04	66.06	3.974
2013	75	468	35.12	63.56	3.927
2014	75	463	36.11	66.33	4.155

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	484	0.384	19.78
Q2	467	1.985	56.04
Q3	476	3.389	111.5
Q4	475	4.952	293.9
Q5	476	6.519	660.1

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
direct_ties	-0.0062	0.0054	-1.159	0.25
component_size	-0.0040	0.0021	-1.878	0.060
open_science	-0.0532	0.0826	-0.645	0.52
log_pat_stock	0.7644	0.0758	10.09	¡ .001
log_assets	0.2436	0.0714	3.413	¡ .001
rd_intensity	4.097	1.240	3.304	¡ .001
direct_ties:open_science	-0.0007	0.0007	-0.887	0.38
component_size:open_science	0.0008	0.0006	1.326	0.18

Full coefficient dump exported from the local estimation script.

model	n	logLik	AIC	BIC	pseudo_r2	r2	within_r2
knowledge_networks	2,276	-25,709	51,788	52,848	0.953	NA	NA

Model fit statistics from the local estimation output.

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